

Engineering Graphics

Subject Code 1005 lesson HTML based on the official SITTTTR Revision 2021 syllabus. It follows a drawing-sheet style with instruments, BIS conventions, dimensioning, engineering curves, projection methods, orthographic views, sectional views, isometric views and CADD basics.

Author: Nandakumar.M | Visit: nandakumarm.dpdns.org

1005 - Engineering Graphics

Engineering Graphics is the language used by engineers to represent shape, size, position and construction details clearly.

1005

Course Code

Engineering Graphics

1

Semester

Diploma in Engineering and Technology

1.5

Credits

45 periods, 3 practical periods per week

Course Category

Engineering Science

Subject Type

Drawing / Graphics subject. So this lesson uses drawing-sheet style, construction steps, labelled visual blocks, drawing rules and answer hints only in the Master Answer Key.

Malayalam Support

Engineering Graphics engineers-ന്റെ drawing language ആണ്. Shape, size, location, construction details എന്നിവ standard drawing methods ഉപയോഗിച്ച് clear ആയി communicate ചെയ്യാൻ ഇത് സഹായിക്കുന്നു. Technical terms English-ൽ തന്നെ retain ചെയ്തിട്ടുണ്ട്.

Course Objectives

- Understand the language of graphics used to express engineering ideas and instructions.
 - Familiarize drawing instruments, drafting and sketching skills.
 - Use Indian Standards related to engineering drawings.
 - Draw and read engineering curves, projections and dimensioning styles.
-
- Develop projection of points and lines.
 - Understand orthographic and sectional views of objects.
 - Visualize actual objects from drawings.
 - Experiment with simple CADD commands for two-dimensional plane figures.

Course Outcomes and Module Map

CO	Outcome	Duration	Cognitive Level
CO1	Illustrate basic elements of Drawing	14 hours	Understanding
CO2	Construct Projections of points and lines	12 hours	Applying
CO3	Build Orthographic projections and Sectional views of object	9 hours	Applying
CO4	Develop Isometric Projections	8 hours	Applying
Series Tests	Series Test I and II	2 hours	Exam practice

Module 1
Drawing basics

Module 2
Points & Lines

Module 3
Orthographic & Section

Module 4
Isometric & CADD

How to Study This Subject

Drawing Practice Method

1. Read the construction aim.
2. Draw a light reference line first.
3. Use correct line type and dimensioning rule.
4. Keep sheet neat; avoid overwriting.
5. Check projection relation before darkening final views.

Exam Preparation Method

1. Revise BIS line types, dimensioning, scale names and projection rules.
2. Practise one polygon, one ellipse/parabola, one point/line projection and one orthographic conversion.
3. For 8-mark and 15-mark drawing questions, write small labels and keep construction visible.
4. Use the Master Answer Key only after trying the expected questions.

Tools, Rules, Symbols and Sheet Memory

Use this bank before entering modules. It replaces formula bank for this graphics subject.

Drawing Instruments

Tool	Use
Drawing board	Supports sheet and gives flat drawing surface.
Mini drafter / T-square	Draws horizontal, vertical and inclined lines.
Set squares	Draws 30°, 45°, 60°, 90° lines.
Compass	Draws arcs and circles.
Divider	Transfers distances accurately.
Scale	Measures and draws required length.

Malayalam Note

Drawing instruments correct ആയി use ചെയ്താൽ drawing neat, accurate, standard ആയിരിക്കും. Pencil pressure light construction-നും dark final line-നും different ആയിരിക്കണം.

Line Types and Dimensioning Rules

Common Engineering Line Types



Continuous thick - visible outline



Dashed - hidden edge



Chain - centre line



Dimension line with arrowheads

Line meaning must be clear before drawing views.

Dimensioning Rules

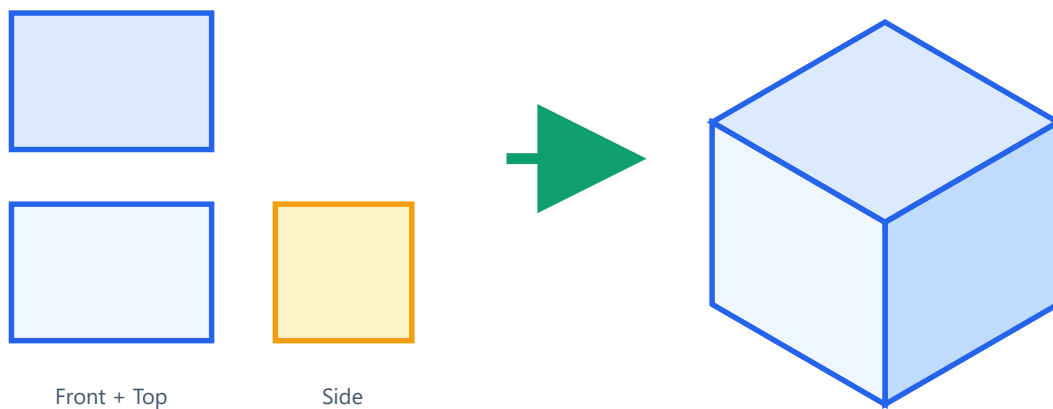
- ✓ Show only necessary dimensions.
- ✓ Place dimensions outside the view where possible.
- ✓ Do not repeat the same dimension unnecessarily.
- ✓ Use arrowheads, extension lines and dimension values neatly.
- ✓ Chain, parallel and coordinate dimensioning are important methods.

Projection and View Bank

Projection Memory

- **HP:** Horizontal Plane.
- **VP:** Vertical Plane.
- **XY line:** reference line between HP and VP.
- **First Angle Projection:** used for exam drawings here.
- **Orthographic views:** front view, top view and side view.
- **Sectional view:** view after imaginary cutting by section plane.
- **Isometric view:** pictorial view with three axes at 120°.

Orthographic to Isometric Idea



Drawing Instruments, Standards, Dimensioning, Curves and Scales

Official focus: basic elements of drawing, instruments, standards, dimensioning, geometrical constructions, engineering curves and scale types.

Learning Outcomes

- Outline the importance of Engineering Graphics.
- Recognize drawing instruments, standards and symbols.
- Apply BIS dimensioning practices.
- Construct regular polygons, conic sections and engineering curves.
- Identify plain, diagonal and vernier scales.

Malayalam Support

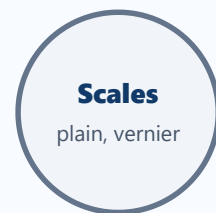
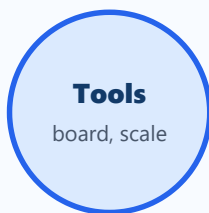
ഈ module drawing-ന്റെ foundation ആണ്. Instruments, line conventions, dimensioning, polygon construction, ellipse, parabola, helix, involute, scale types എന്നിവ neat ആയി practise ചെയ്യണം.

Simple Explanation and Important Definitions

Core Ideas

- **Engineering Graphics:** Standard drawing method used to communicate engineering objects.
- **Convention of lines:** Standard line types used for visible edges, hidden edges, centre lines, dimensions and cutting planes.
- **Dimensioning:** Method of showing sizes and locations on a drawing.
- **Conic sections:** Curves obtained by cutting a cone, such as ellipse and parabola.
- **Engineering curves:** Curves such as helix and involute used in springs, threads, gears and machine parts.

Module 1 Visual Map



Construction Procedure Cards

Regular Polygon

1. Draw given side length.
2. Locate centre or use angle method.
3. Mark all vertices accurately.
4. Join vertices with continuous thick lines.

Ellipse

1. Draw major and minor axes.
2. Use eccentricity / concentric circle / rectangular method as asked.
3. Plot points symmetrically.
4. Join points with smooth curve.

Parabola

1. Draw axis, directrix and focus if needed.
2. Use eccentricity or tangent method.
3. Locate equal-distance points.
4. Draw smooth parabola through points.

Expected Questions - Module 1

- 1** M1Q1. Explain the importance of Engineering Graphics in engineering practice.
- 2** M1Q2. List the main drawing instruments and write their uses.
- 3** M1Q3. Explain common line conventions used in engineering drawing.
- 4** M1Q4. Define dimensioning and compare chain, parallel and coordinate dimensioning.
- 5** M1Q5. Write the construction steps for a regular polygon when the side length is given.
- 6** M1Q6. Explain any two methods used for constructing an ellipse.
- 7** M1Q7. Write short notes on helix and involute with applications.
- 8** M1Q8. Identify plain scale, diagonal scale and vernier scale and state where each is useful.

Projection of Points and Lines

Official focus: theory of projection, projection of points in different quadrants and projection of straight lines in first quadrant.

Learning Outcomes

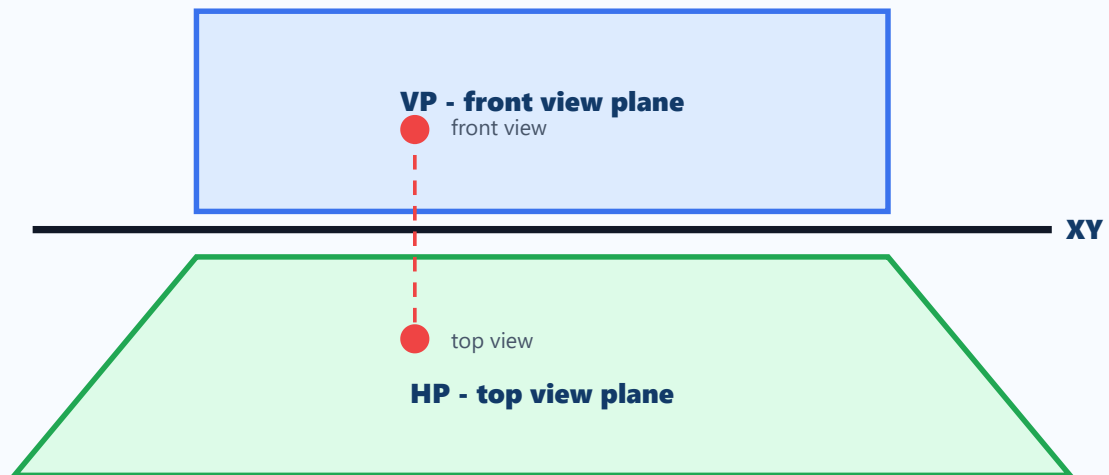
- Outline the theory of projection.
- Develop projections of points in different quadrants.
- Develop projections of lines parallel or perpendicular to reference planes.
- Draw line projections inclined to one or both planes.
- Find true length and inclination with HP and VP.

Malayalam Support

Projection-ൽ object-ന്റെ points അല്ലെങ്കിൽ lines HP, VP എന്നീ reference planes-ലേക്ക് project ചെയ്യുന്നു. XY line, front view, top view, quadrant position എന്നിവ carefully mark ചെയ്യണം.

Projection Theory Visual

HP, VP and XY Reference



Always keep vertical projectors light and aligned.

Important Definitions

- **Projection:** Representation of an object on a plane using projectors.
- **Projector:** Imaginary line from object to projection plane.
- **Front view:** Projection on VP.
- **Top view:** Projection on HP.
- **True length:** Actual length of a line, seen when line is parallel to the projection plane.

Line Position Quick Table

Line Position	View Behaviour	Drawing Note
Parallel to both HP and VP	Both views show true length.	Keep views parallel to XY.
Parallel to one plane and perpendicular to other	One view true length, other may appear as point/end view.	Check plane relation before drawing.
Inclined to one plane and parallel to other	One view true length, other view shortened.	Angle is visible in plane where true length appears.
Inclined to both HP and VP	Both views appear shorter than true length.	Use rotation / auxiliary method as instructed.

Expected Questions - Module 2

- 1** M2Q1. Explain the theory of projection using HP, VP and XY line.
- 2** M2Q2. Draw and explain projection of a point placed in different quadrants.
- 3** M2Q3. State the difference between front view and top view.
- 4** M2Q4. Draw projections of a straight line parallel to both HP and VP.
- 5** M2Q5. Draw projections of a line parallel to one plane and perpendicular to the other plane.
- 6** M2Q6. Explain the procedure for a line inclined to one plane and parallel to the other plane.
- 7** M2Q7. Explain how true length and inclination of a line are found from projections.
- 8** M2Q8. List common mistakes made while drawing projections of points and lines.

Orthographic Projection and Sectional Views

Official focus: orthographic projection, first angle and third angle methods, conversion of pictorial views into orthographic views and full/half sectional views.

Learning Outcomes

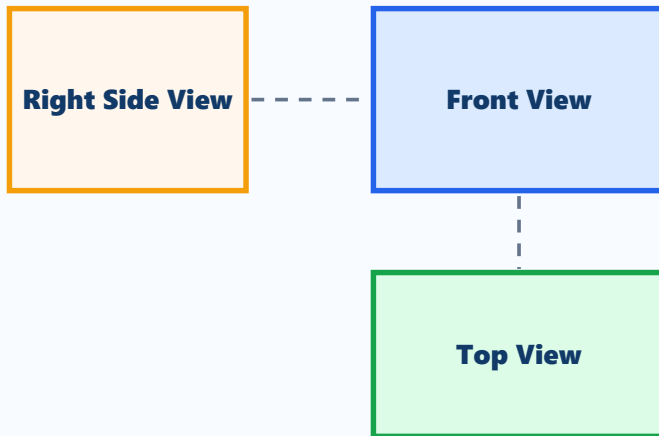
- Summarize orthographic, isometric and oblique projections.
- Use first angle and third angle projection symbols.
- Convert pictorial views into orthographic views.
- Draw objects with plain, slanting, slotted, ribbed and cylindrical surfaces.
- Develop full and half sectional views.

Malayalam Support

Orthographic projection-ൽ object-നെ front, top, side views ആയി കാണിക്കുന്നു. ഈ syllabus-ൽ exam drawing first angle projection method ഉപയോഗിച്ചാണ് ചെയ്യേണ്ടത്.

Orthographic View Layout

First Angle Projection Layout Idea



In first angle, top view is usually placed below front view.

Sectional View Need

Sectional views reveal internal details hidden by outside surfaces. A cutting plane is imagined through the object, the front portion is removed, and the remaining cut face is shown using hatching.

Internal details കാണിക്കാൻ object cut ചെയ്ത പോലെ കാണിക്കുന്ന drawing ആണ് sectional view. Cut surface-ൽ hatching lines വരയ്ക്കണം.

Projection Comparison

Projection Type	Purpose	Important Note
Orthographic	Shows exact front, top and side views.	Best for manufacturing and inspection.
Isometric	Shows pictorial 3D-like appearance.	Three axes are equally inclined.
Oblique	Shows front face true shape with depth drawn at an angle.	Concept and applications only here.
Sectional	Shows internal details.	Use correct section plane and hatching.

Expected Questions - Module 3

- 1** M3Q1. Define orthographic projection and state its engineering importance.
- 2** M3Q2. Compare first angle projection and third angle projection.
- 3** M3Q3. Draw the standard symbol of first angle projection and explain its meaning.
- 4** M3Q4. Convert a simple pictorial object into front view, top view and side view using first angle projection.
- 5** M3Q5. Explain how slanting surfaces, slots, ribs and cylindrical surfaces are represented in orthographic views.
- 6** M3Q6. Explain the need for sectional views in engineering drawing.
- 7** M3Q7. Differentiate full section and half section with suitable drawing situations.
- 8** M3Q8. List the common mistakes to avoid while preparing orthographic and sectional drawings.

Isometric Projection, Freehand Sketching and CADD Basics

Official focus: isometric scale, isometric view/projection, conversion between orthographic and isometric views, freehand sketching and CADD familiarization.

Learning Outcomes

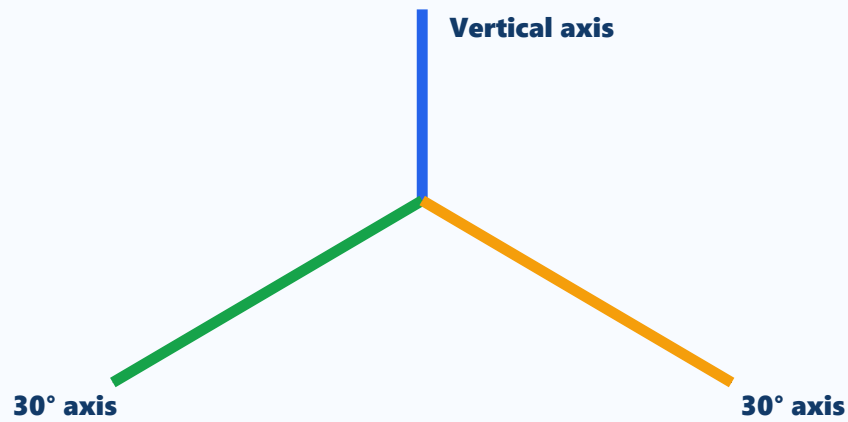
- Develop isometric views of simple engineering objects.
- Convert orthographic views into isometric views/projections.
- Prepare freehand sketches of real objects.
- Convert isometric view to orthographic view and vice versa.
- Identify basic CADD commands for two-dimensional figures.

Malayalam Support

Isometric drawing object-നെ 3D-like ആയി കാണിക്കാൻ ഉപയോഗിക്കുന്നു. Orthographic views-ൽ നിന്ന് height, width, depth identify ചെയ്ത് isometric axes-ൽ transfer ചെയ്യുന്നു.

Isometric Axes and Conversion

Isometric Axes



The three isometric axes are 120° apart.

Isometric Terms

- **Isometric view:** Drawn using true lengths along isometric axes.
- **Isometric projection:** Drawn using isometric scale, so size is reduced.
- **Isometric scale:** Scale used to convert true length to isometric length.
- **Freehand sketch:** Drawing without instruments, using proportion and observation.

CADD Familiarization

Software Mentioned in Syllabus

- AutoCAD
- Electrical CAD
- STAAD
- Mechanical Desktop

Basic 2D Commands

LINE

CIRCLE

ARC

OFFSET

TRIM

EXTEND

MOVE

COPY

DIMENSION

CADD commands drawing speed കൂട്ടുകയും accuracy improve ചെയ്യുകയും ചെയ്യുന്നു. എന്നാൽ basic manual drawing concepts അറിയണം.

Expected Questions - Module 4

- 1** M4Q1. Define isometric view, isometric projection and isometric scale.
- 2** M4Q2. Draw the isometric axes and explain the angle relation between them.
- 3** M4Q3. Convert given orthographic views of a simple object into an isometric view.
- 4** M4Q4. Explain the procedure for converting an isometric view into orthographic views.
- 5** M4Q5. Write the importance of freehand sketching in engineering practice.
- 6** M4Q6. Prepare a freehand isometric sketch of a simple real object such as a block, bracket or step block.
- 7** M4Q7. List the software packages introduced under CADD familiarization.
- 8** M4Q8. Identify any six basic CADD commands used for constructing two-dimensional plane figures.

One-page Last-minute Revision Sheet

Use this for final checking before drawing practice or exam.

Important Definitions

- Engineering Graphics: standard graphical language of engineering.
- Dimensioning: method of showing size and location.
- Projection: representation of an object on a plane.
- Orthographic projection: exact front/top/side view method.
- Sectional view: view showing internal details by cutting the object.
- Isometric view: pictorial view along three isometric axes.

Important Procedures

- Draw construction lines light and final object lines dark.
- Keep all projectors aligned in projection problems.
- Use first angle projection for required exam drawings.
- Place top view below front view in first angle layout.
- Use hatching only on cut surfaces in sectional views.
- Transfer width, height and depth carefully in isometric drawing.

Common Exam Mistakes

- ✗ Using wrong line type for hidden edge or centre line.
- ✗ Writing dimensions inside the object unnecessarily.
- ✗ Confusing first angle and third angle view positions.
- ✗ Not aligning front view and top view with projectors.
- ✗ Darkening construction lines too early.
- ✗ Forgetting hatching in sectional view.
- ✗ Using true scale where isometric scale is specifically required.

Malayalam Checklist

- ✓ Sheet clean ആണോ?
- ✓ Line type correct ആണോ?
- ✓ Dimension repeated അല്ലേ?
- ✓ XY line clear ആണോ?
- ✓ Front, top, side views aligned ആണോ?
- ✓ Hatching correct direction and spacing ആണോ?

Master Answer Key

Answers / Hints Matched to Expected Questions

Every answer below matches the same question number shown inside Module 1 to Module 4.
No extra answer is added.

Module 1 Answer Key

M1Q1

Engineering Graphics is important because it communicates engineering shape, size, location, construction details and manufacturing instructions in a standard visual form. It avoids misunderstanding between designer, technician, fabricator and inspector. Add examples: machine part drawing, civil plan, electrical panel layout.

M1Q2

Drawing board: support sheet; mini drafter/T-square: horizontal and vertical lines; set squares: 30°, 45°, 60°, 90° lines; compass: circles and arcs; divider: transfer distance; scale: measure length; pencil: construction and final lines; eraser/sharpener: correction and preparation.

M1Q3

Continuous thick line shows visible outlines. Continuous thin line is used for construction/dimension/extension lines. Dashed line shows hidden edges. Chain thin line shows centre lines. Cutting plane line indicates section plane. Use a small labelled line sample in the answer.

M1Q4

Dimensioning shows size and position of features using dimension lines, extension lines, arrowheads and values. Chain dimensioning places dimensions one after another. Parallel dimensioning gives dimensions from a common base line. Coordinate dimensioning gives positions using coordinates from a datum.

M1Q5

Construction steps: draw the given side; choose suitable polygon construction method; locate centre/vertices accurately; mark all equal sides; join adjacent vertices; darken final

polygon; add labels and required dimensions. Mention neat construction and equal side checking.

M1Q6

Ellipse may be constructed by eccentricity method, concentric circle method or rectangular method. In concentric circle method, draw major/minor axes, draw two concentric circles, divide circles equally, project points and draw a smooth curve. In rectangular method, draw enclosing rectangle, divide sides equally and obtain points by intersection.

M1Q7

Helix is a curve generated by a point moving around a cylinder/cone while also moving along its axis; applications include screw threads and springs. Involute is generated by unwinding a taut string from a circle or polygon; applications include gear tooth profile and cam mechanisms.

M1Q8

Plain scale measures two units such as metre and decimetre. Diagonal scale measures three units more accurately, such as metre, decimetre and centimetre. Vernier scale gives fine measurement using vernier principle. In this syllabus, construction is not required; identification and purpose are important.

Module 2 Answer Key

M2Q1

Projection is obtained by drawing projectors from an object to a projection plane. VP gives front view and HP gives top view. The intersection of HP and VP is represented by XY line. The object position with respect to HP and VP decides the quadrant and view locations.

M2Q2

In first quadrant, point is above HP and in front of VP; front view is above XY and top view below XY. In second quadrant, point is above HP and behind VP. In third quadrant, point is below HP and behind VP. In fourth quadrant, point is below HP and in front of VP. Draw XY line and mark views according to distances.

M2Q3

Front view is projection on VP and shows height relation. Top view is projection on HP and shows plan/depth relation. Front view and top view should be aligned with the same projector.

M2Q4

When a line is parallel to both HP and VP, both front view and top view show true length. Draw XY line, locate end projections, keep the two views parallel to XY, mark equal true lengths and label endpoints correctly.

M2Q5

For a line parallel to one plane and perpendicular to the other, the plane to which it is parallel shows true length. The plane to which it is perpendicular shows an end view or reduced/point representation depending on position. Use projectors from both endpoints and label views.

M2Q6

Procedure: draw reference XY; draw the view parallel to the plane where the line remains parallel; mark given inclination in the other view; project endpoints; transfer distances from HP/VP; complete corresponding view; label apparent length and true length if applicable.

M2Q7

True length is seen when the line is parallel to the projection plane. If the line is inclined to both planes, use rotation or auxiliary construction to place one view parallel to XY/plane. The angle made by the true-length view with reference gives inclination with HP or VP as required.

M2Q8

Common mistakes: wrong quadrant placement, missing XY line, unaligned projectors, confusing front view and top view, wrong distance transfer from HP/VP, not labelling endpoints, using dark lines for construction, and not showing true length clearly.

Module 3 Answer Key

M3Q1

Orthographic projection represents an object using separate views on mutually perpendicular planes, normally front, top and side views. It is important because it shows true shape and size needed for manufacturing, assembly and inspection.

M3Q2

In first angle projection, the object is imagined between observer and plane; top view is placed below front view and right side view is placed on the left side. In third angle projection, the plane is between observer and object; top view is placed above front view and right side view on the right side.

M3Q3

The first angle symbol uses the frustum of a cone with its circular end view placed on the left of the frustum view. Explain that the symbol identifies the projection system used in the drawing sheet. Draw the symbol neatly and label it as First Angle Projection.

M3Q4

Procedure: study pictorial object; identify front direction; draw front view first; project downward to get top view; project sideward to get side view; transfer depths using 45° mitre line if needed; draw visible outlines thick, hidden lines dashed and centre lines chain; check alignment.

M3Q5

Slanting surfaces appear as inclined edges or foreshortened surfaces in related views. Slots are shown by visible or hidden outlines depending on location. Ribs are represented by

their outline and section convention. Cylindrical surfaces require centre lines and circular/rectangular appearance depending on viewing direction.

M3Q6

Sectional views are needed to show internal holes, cavities, slots and hidden construction clearly. Instead of using many hidden lines, an imaginary cutting plane cuts the object and the cut surface is hatched. It improves clarity for manufacturing and inspection.

M3Q7

Full section cuts the object completely through a cutting plane and shows the entire internal section. It is used for symmetrical or complex objects where full internal details are required. Half section shows one half outside view and one half sectional view, often used for symmetrical objects.

M3Q8

Avoid wrong view placement, missing projectors, mismatched sizes between views, wrong hidden-line use, missing centre lines, incorrect hatching, hatching non-cut areas, poor selection of cutting plane, and mixing first angle and third angle conventions.

Module 4 Answer Key

M4Q1

Isometric view is a pictorial drawing made using true lengths along isometric axes. Isometric projection is made using isometric scale, so lengths are reduced. Isometric scale converts true length to isometric length for accurate isometric projection.

M4Q2

Draw one vertical axis and two axes inclined at 30° to the horizontal on both sides. The three isometric axes are 120° apart. Width, height and depth are transferred along these axes.

M4Q3

Procedure: read front, top and side views; identify overall length, width and height; draw isometric axes; make an isometric box; mark cuts, steps, holes or slots; complete visible edges; remove unnecessary construction; darken final object lines.

M4Q4

To convert isometric to orthographic, choose front direction, read height, width and depth from isometric view, draw front view, project top view and side view, align all views, represent hidden edges if needed and add centre lines for cylindrical features.

M4Q5

Freehand sketching is useful for quick communication of ideas, site notes, workshop explanation, design thinking and rough planning. It improves visualization and helps before preparing accurate instrument/CADD drawings.

M4Q6

Hint: Choose a simple object such as a step block. Draw three isometric axes lightly, mark approximate proportions, draw main box, add step/cut details, keep vertical lines vertical and receding lines at about 30°, then darken final visible edges.

M4Q7

Software packages introduced: AutoCAD, Electrical CAD, STAAD and Mechanical Desktop. Their purpose is to prepare accurate computer-aided drawings/models and improve drafting productivity.

M4Q8

Any six basic CADD commands: LINE, CIRCLE, ARC, RECTANGLE, OFFSET, TRIM, EXTEND, MOVE, COPY, MIRROR, ROTATE, FILLET, CHAMFER, DIMENSION. State that these commands help construct and edit 2D plane figures.

Author: Nandakumar.M | **Visit:** nandakumarm.dpdns.org